MIL-D-60362A (PA)
7 May 1975
Superseding
MIL-D-60362 (MU)
11 February 1966 and
MIL-D-60360 (MU)
11 February 1966
(see 6.6)

MILITARY SPECIFICATION

DETONATOR, STAB, M42 PARTS FOR AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of stab detonator designated as Detonator, Stab, M42.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
*MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspections by Attributes (ABC-STD-105)
MIL-STD-650 - Explosives; Sampling, Inspection and Testing

FSC: 1375

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

*(For meaning of asterisk (*) see paragraph 6.5)

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

*PRODUCT AND PACKING DRAWINGS

9297975 - Disc, Detonator
9297976 - Disc, Detonator, Closing
9297977 - Cup Detonator
8797787 - Detonator, Stab, M42, Assembly
9258182 - Box, Set-Up, Packing, Ammunition, Inner for Small Explosive Components
9258183 - Box, Set-Up, Packing, Ammunition, Outer, for Small Explosive Components
9258184 - Box, Packing, Ammunition for Small Explosive Components

*INSPECTION EQUIPMENT DRAWINGS

9200695 - Firing Assembly
8827934 - Adapter, Ball
9200012 - Pin, Firing
8816367 - Guide, Firing Pin
9200602 - Bushing
8816368 - Disc, Lead

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation E300 - Industrial Chemicals, Recommended Practice for Sampling

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn. 19103).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. - The parts shall comply with all requirements specified on Drawings (dwgs. 9297975, 9297976 and 9297977 as applicable and with all requirements specified in applicable specifications.

3.3 Detonator. - The detonator shall comply with all requirements specified on Drawing (dwg.) 8797787 and with all requirements specified in applicable specifications.

3.4 Moisture content of lead azide. - The moisture content of the lead azide used in loading the detonator shall not exceed 0.50 percent.

3.5 Moisture content of primer mixture. - The moisture content of the primer mixture used in loading the detonator shall not exceed 0.30 percent.

3.6 Functioning

3.6.1 Output. - The detonator shall function on the first attempt and shall cause perforation of a .1345 inch thick lead disc, to a diameter of not less than 1/4 inch.

3.6.2 Waterproofness. - The detonators shall be waterproof as evidenced by normal functioning, as specified in 3.6.1 after being immersed in water maintained at 75 degrees Fahrenheit (F.) plus or minus 5 degrees F., at a depth of 2 inches minimum, for not less than 48 hours.

*3.7 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used

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shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

3.8 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

*4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

*4.3.1 Submission. - The contractor shall submit a first article sample (see 6.1b) as designated by the Contracting Officer. The first article sample shall consist of fifty (50) complete sets of parts and subassemblies (i.e. 50 each of every component and sub-assembly) and 340 loaded detonators. The first article shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

*4.3.2 Inspections to be performed. - See MIL-A-48078 and paragraphs 4.4.2.1 through 4.4.2.3 and Table I specified herein.

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TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60362A (PA)

PARAGRAPH *	TITLE: Detonator, Stab, M42, Metal Parts and Assemblies		SHEET 1 OF 1		DRAWING NUMBER: See Below; NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Metal Parts Dwg. 9297975, 9297976 and 9297977 Examination for defects	50(a)		3.2	4.4.2.1 through 4.4.2.3
	Detonator Assemblies Dwg. 8797787 Examination for defects	50(b)		3.3	4.4.2.5
	Testing Functioning Waterproofness	315 25		3.3	4.4.3.4.1 4.4.3.4.2
Notes	(a) 50 each and every component and sub-assembly (b) From 340 prior to testing				

NOTES:

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*4.3.3 Rejection. - (See MIL-A-48078)

4.4 Quality conformance inspection

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

c. Primer mixture from one batch or blending of batches which results in homogeniety of all primer mixture used in loading the entire detonator lot.

d. Tetryl from not more than one lot.

*4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60362A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.1	Disc, Detonator				9297975
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Diameter		0.40%	3.2	Gage
132	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60362A (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.2	Disc, Detonator Closing		9297976
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Diameter		0.40%	3.2	Gage
132	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60362A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.3	Cup, Detonator				9297977
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup, split or cracked		100.0%	3.2	Visual
Major B					
131	Diameter of flash hole, min.		0.40%	3.2	Gage
132	Concentricity of flash hole with outside diameter		0.40%	3.2	Gage
133	Thickness through bottom		0.40%	3.2	Gage
134	Length		0.40%	3.2	Gage
135	Outside diameter, max		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60362A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.4	Assembly, Prior to Loading				8797787
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Interior coating incomplete or missing		100.0%	3.3	Visual
Major A&B	None defined				
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60362A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8797787
*4.4.2.5	Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup, cracked or split		100.0%	3.3	Visual
Major B					
131	Length		0.40%	3.3	Gage
132	Diameter		0.40%	3.3	Gage
133	Disc loose, punctured, torn or wrinkled to such a degree that the contents are exposed		0.40%	3.3	Visual
134	Color of lacquer incorrect on either end		0.40%	3.3	Visual
135	Cup damaged except as described in critical above		0.40%	3.3	Visual
136	Crimp not full 360 degrees		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60362A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.6	Box, Set-Up, Inner, After Loading and Prior to Closing Slide Cover				9258182
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Filler and cushion assembly missing		100.0%	3.3	Visual
2	Filler and cushion assembly improperly assembled		100.0%	3.3	Visual
3	Stiffner missing		100.0%	3.3	Visual
4	Slide cover assembled reversed		100.0%	3.3	Visual
Major A					
101	Release paper missing		0.40%	3.3	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.3	Visual
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60362A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.7	Box, Set-Up, Outer, with Inner Set-Up Boxes Assembled and Prior to Assembling Top				9258183
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Red surface on inner set-up boxes does not alternate on same side of outer set-up boxes		100.0%	3.3	Visual
2	Stiffner not facing one direction		100.0%	3.3	Visual
Major A					
101	Inner box does not fit properly in outer box		0.40%	3.3	Visual
102	Fillers missing		0.40%	3.3	Visual
Major B					
131	Label not securely applied to box		0.40%	3.3	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.3	Visual
202	Box improperly taped		0.65%	3.3	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60362A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.8	Sealed Packed Barrier Bag, Prior to Packing in Box				9258183
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Waterproofness of barrier bag destroyed by rupture, puncture or separation of heat seal		0.40%	3.3	Visual
102	Evidence of crushed box		0.40%	3.3	Visual
103	All excess air not exhausted from bag prior to sealing		0.40%	3.3	Visual
NOTE:	Package should be opened if there is evidence of crushing				
Minor					
201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.3	Visual
202	Marking of bag misleading or unidentifiable		0.65%	3.3	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60362A, (PA)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
*4.4.2.9	Box, Packing, Ammunition Prior to Assembling Cover and Strapping	1	1	9258184
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Bags containing set-up boxes damaged		0.40%	3.3	Visual
102	Cushioning material incomplete or missing from around outer set-up boxes		0.40%	3.3	Visual
103	Marking missing, misleading or unidentifiable		0.40%	3.3	Visual
104	Bottom cushioning missing or of incorrect thickness		0.40%	3.3	Visual
Minor					
201	Contents loose within box		0.65%	3.3	Visual
202	Evidence of poor workmanship		0.65%	3.7	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-60362A(PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.10	Box, Packing, Ammunition		9258184
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Strapping missing, broken, loose or improperly fastened		0.40%	3.3	Visual/Manual
102	Box damaged		0.40%	3.3	Visual
103	Marking missing, incomplete or illegible		0.40%	3.3	Visual
Minor					
201	Wood box cover nailed		0.65%	3.3	Visual
202	Metallic seal missing		0.65%	3.3	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

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4.4.3 Testing (See Note)

Note: Chemical Sampling. - A representative sample shall be randomly selected for the applicable tests specified in 4.4.3. The procedure described in ASTM E300 shall be used to withdraw the samples. The applicable tests specified in 4.5 shall be performed using prescribed analytical procedure for replicate determination given in standard analytical textbooks.

4.4.3.1 Binder-lubricant content of tetryl (see dwg. 8797787) - Major A defect. - The binder-lubricant content of the tetryl shall be determined for each lot at the time of loading. A sample, in sufficient quantity to perform the test detailed in 4.5.1, shall be selected and tested as specified in 4.5.1. If the binder-lubricant content is in excess of the percentage specified and if loading has not begun, the lot of tetryl shall be rejected. If loading has begun, action on the unloaded tetryl shall be as specified above and all detonators loaded with questionable tetryl shall be rejected.

4.4.3.2 Moisture content of lead azide (see 3.4) - Major A defect. - The moisture content of the lead azide shall be determined for each lot prior to preparing the primer mixture composition. A sample in sufficient quantity to perform the test detailed in 4.5.2 shall be selected and tested as specified in 4.5.2. If the moisture content is in excess of the percentage specified and loading has not begun the lot of lead azide shall be rejected. If loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded with the questionable lead azide shall be rejected.

4.4.3.3 Primer Mixture

4.4.3.3.1 Moisture content (see 3.5) - Major A defect. - The moisture content of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample, in sufficient quantity to perform the test detailed in 4.5.3.1 shall be selected and tested as specified in 4.5.3.1. If the moisture content is in excess of the percentage specified and loading has not begun, the batch or blend of batches of primer mixture shall be rejected. If loading has begun, action on the unloaded primer mixture shall be as specified above and all detonators loaded with the questionable primer mixture shall be rejected.

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4.4.3.3.2 Composition (see dwg. 8797787) - Major A defect. - The composition of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample, in sufficient quantity to perform the test detailed in 4.5.3.2, shall be selected and tested as specified in 4.5.3.2. If the composition of the primer mixture fails to conform to the requirements specified on the applicable drawing and if loading has not begun, the batch or blend of batches of primer mixture shall be rejected. If loading has begun, action on the unloaded primer mixture shall be as specified above and all detonators loaded with the questionable primer mixture shall be rejected.

4.4.3.4 Functioning

4.4.3.4.1 Functioning (see 3.6.1) - Major A defect. - A sample of 315 detonators shall be selected at random from each lot. If 3 or more detonators fail to comply with the requirement specified the lot shall be rejected. If one or two defects are found, a second sample of 315 shall be selected at random and tested. If, in the combined first and second sample, a total of 4 or more defects are found, the lot shall be rejected. The test shall be performed as specified in 4.5.4.1 using test equipment as specified in 4.4.4.

4.4.3.4.2 Waterproofness (see 3.6.2) - Major A defect. - A sample of 25 detonators shall be selected at random from each lot. If one or more detonators fail to comply with the requirement specified the lot shall be rejected. The test shall be performed as specified in 4.5.4.2 using test equipment as specified in 4.4.4.

4.4.3.5 Heat seal test of sealed bag (see dwg. 9258183) Major A defect. - Sampling, acceptance criteria and test procedures for this test shall be as specified in Specification MIL-P-116.

4.4.4 Inspection Equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and section 6.2 herein.

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4.5 Test methods and procedures

4.5.1 Determination of binder-lubricant content of tetryl. - An accurately weighed portion of 5 grams shall be transferred to a tared medium porosity sintered glass crucible. The sample shall be extracted with four 10 milliliter (ml.) portions of acetone, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction. Any lumps which may persist shall be broken up and the sides of the crucible washed down with each 10 ml. portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 95 degrees Centigrade plus or minus 5 degrees C. for 1/2 hour, cooled in a desiccator and weighed. The above extraction shall be repeated, drying and weighing with individual 10 ml. portions of acetone until the loss in weight on subsequent extractions totals less than 0.005 gram. The weight of residue in the crucible shall be calculated as the percentage of total binder-lubricant.

4.5.2 Moisture content of lead azide. - The moisture content of the lead azide shall be determined in accordance with MIL-STD-650 Method 101.5 except, use a two gram sample and cover the dish with a sheet of polytetrafluoroethylene.

4.5.3 Primer mixture

4.5.3.1 Moisture content of primer mixture. - An accurately weighed 0.7 gram portion of the sample shall be dried to constant weight in an oven, maintained at 60 degrees C. plus or minus 5 degrees C., or in a desiccator, containing anhydrous calcium chloride. The loss in weight shall be calculated as the percentage of moisture.

4.5.3.2 Composition

4.5.3.2.1 Potassium chlorate

4.5.3.2.1.1 Preparation of solutions

4.5.3.2.1.1.1 Solution Number 1. - To approximately 800 ml. distilled water, in a one liter amber glass reagent bottle, 40.0 grams (gm.) of potassium thiocyanate shall be added. The bottle shall be sealed with a rubber stopper and the reagent

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bottle and contents shaken until the solid material is dissolved. To this solution 1.0 gm. of lead thiocyanate, 1.0 gm. of antimony sulfide and 1.0 gm. of lead azide shall be added. The bottle shall again be stoppered and the reagent bottle and contents vigorously shaken for several minutes, and then allowed to stand overnight. Portions of the saturated solution shall be filtered before using.

4.5.3.2.1.1.2 Solution Number 2. - To approximately 400 ml. of distilled water, in a one-half liter amber glass reagent bottle, 1.0 gm. of lead thiocyanate, 1.0 gm. of antimony sulfide and 1.0 gm. of lead azide shall be added. The bottle shall be sealed with a rubber stopper and the reagent bottle and contents vigorously shaken for several minutes and then allowed to stand overnight. Portions of the saturated solution shall be filtered before using.

4.5.3.2.1.2 Procedure. - A portion of approximately 1.0 gm. of the sample shall be accurately weighed directly into a tared 30 ml. sintered glass crucible of medium porosity. From a burette 3.0 ml. of Solution Number 1 shall be added to the crucible. The solution shall be swirled for exactly one minute, taking care not to spill any liquid; aspirated with suction and the bottom of the crucible wiped dry, using tissue paper. The washing shall be continued, using one more 3.0 ml. portion, two 2.0 ml. and two 1.0 ml. portions, making a total of 12.0 ml. The crucible shall be removed and the bottom of the crucible, the funnel, the adapter and the suction flask washed with water. The test tube shall be placed in the flask and the stem of the filtering funnel inserted into the test tube. The crucible and contents shall be washed with one 1.0 ml. portion of Solution Number 1 as above, and aspirated with suction, catching the filtrate in the test tube. The test tube shall be removed, 1.0 ml. of distilled water added and the chlorate ion, determined as follows.

The tube shall be inclined at an angle of about 45 degrees and from a dropper about 0.5 ml. of a solution containing 1.0 gm. of diphenylamine in 100 gm. of concentrated sulfuric acid shall be run down the side of the tube, so that there will be two distinct layers. If an appreciable amount of chlorate ion is present, a distinct blue ring will be visible at the border of the two layers. On shaking the tube, the ring disappears, but the contents assume a blue coloration, which lasts several seconds, depending on the amount of chlorate ion present. If only a trace

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of chlorate ion is present the blue ring may not form, but upon shaking the test tube, a slight bluish coloration will appear momentarily throughout the solution, lasting only a fraction of a second. If a definite blue ring forms, the washing shall be repeated as before using one 1.0 ml. portion of Solution Number 1, and finally a 0.5 ml. portion of the same solution, as needed, depending on the outcome of the tests for the chlorate ion. (Usually a total of 14.0 or 14.5 ml. of Solution Number 1 is sufficient). After complete removal of potassium chlorate the inside of the crucible shall be washed with 1.0 ml. of Solution Number 2 and aspirated immediately. The bottom of the crucible shall be wiped dry with tissue paper and dried in an oven at 80 degrees C. plus or minus 1 degree C. for 30 minutes, cooled in a desiccator and weighed. The percentage of potassium chlorate shall be calculated as follows:

$$\text{Potassium Chlorate, percent} = \frac{100B}{W(1-M)}$$

where:

B = loss in weight of contents of crucible, grams.
W = weight of sample, grams.
M = $\frac{\text{percent moisture}}{100}$

4.5.3.2.2 Antimony sulfide

4.5.3.2.2.1 Preparation of solution of Ferric ammonium sulfate. - To approximately 120 ml. of distilled water, in a one-half liter reagent bottle, 125 gm. of ferric ammonium sulfate shall be added. The bottle shall be stoppered and shaken vigorously for several minutes and then allowed to stand overnight. Portions of the saturated solution shall be filtered before using.

4.5.3.2.2.2 Procedure. - A portion of approximately 1 gm. of the sample shall be accurately weighed directly into a tared 30 ml. sintered glass crucible of medium porosity. Five ml. of one percent nitric acid shall be added and the crucible constantly swirled for one minute in order to dissolve all the ingredients of the powder except antimony sulfide. Care shall be taken not to spill any liquid. The mixture shall be filtered by applying suction employing a vacuum flask. The crucible shall be removed from the adapter and the bottom wiped dry with tissue paper. The operations of washing, using 5.0 ml. of one percent nitric acid

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each time and swirling for exactly one minute shall be repeated. After a total of 40 ml. of one percent nitric acid has been used, the crucible shall be removed and the bottom of it washed with water. The funnel, the funnel adapter and the vacuum flash shall be washed with water; the filtrate and washings shall be discarded. About 1 ml. of a saturated solution of ferric ammonium sulfate shall be placed in the flask. The contents of the crucible shall be treated as before with 5.0 ml. of one percent nitric acid and filtered by means of suction into the flask. If the liquid in the flask turns red, due to the presence of the thiocyanate ion in the filtrate, the washing of the contents in the crucible shall be repeated with additional 5.0 ml. portions of one percent nitric acid until a negative test for the thiocyanate ion is affected. (Usually a total of 50 ml. is sufficient). The crucible and contents shall be rinsed with alcohol and then with ether, and then aspirated until the odor of ether is no longer detectable. The crucible and contents shall be dried in an oven at 80 degrees C. plus or minus 1 degree C. for 30 minutes, cooled in a desiccator and weighed. The percentage of antimony sulfide shall be calculated as follows:

$$\text{Antimony sulfide, percent} = \frac{100A}{B}$$

where:
A = weight of residue in the crucible.
B = weight of sample on a moisture-free basis.

4.5.3.2.3 Lead azide. - The lead azide content shall be determined using the (Eudiometer) method and apparatus in MIL-STD-650 Method no. 407.1.

5.4.3.2.4 Lead thiocyanate. - The percentage of lead thiocyanate in the sample shall be calculated on a moisture-free basis by subtracting from 100 the sum of the percentages of potassium chlorate, antimony sulfide and lead azide.

4.5.4 Functioning

4.5.4.1 Output. - The detonator shall be assembled to the applicable test fixture specified in 4.4.4 so that the firing pin will strike the sensitive end of the detonator. A steel ball 1 3/16 inch in diameter and weighing 3.95 ounces maximum, shall be dropped on the firing pin to function the detonator, from a height of 3 inches, maximum. The height of drop shall be

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measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective. Mehtod of inspection shall be in accordance with dwg. 9200695, 8827934, 9200012, 8816367, 9200602, and 8816368.

4.5.4.2 Waterproofness. - The detonator shall be immersed under water as specified for the time specified. Within one hour after removal from the water the detonator shall be subjected to the output test specified in 4.5.4.1. Any detonator which fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

*5.1.1 Level C. - The detonators shall be packaged and marked in accordance with dwg. 9258182 and dwg. 9258183.

*5.1.2 Level C. - The detonator parts shall be packaged in accordance with MIL-STD-1169 requirements.

5.2 Packing

*5.2.1 Level C. - The detonators shall be packed as specified on dwg. 9258184.

*5.2.2 Level C. - The detonator parts shall be packed in accordance with MIL-STD-1169 requirements.

5.3 Detonator containers and bags shall be marked in accordance with instructions contained in dwgs. 9258182, 9258183 and 9258184. Detonator parts containers shall be marked in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Provisions of MIL-A-48078.

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6.2 Submission of Inspection Equipment Designs for Approval. - (See MIL-A-48078). Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T Dover, New Jersey 07801.

6.3 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M Dover, New Jersey 07801.

6.4 Check test for possible deterioration. - The following test is designed to prevent detonators that have deteriorated from being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Detonator assemblies shall be selected from each lot for this test.

6.4.1 Functioning. - A sample of 315 detonator shall be selected at random from each lot. If 4 or more detonators fail to comply with the requirement specified, the lot shall be rejected. If more than one defect is found, a second sample of 315 shall be selected at random and tested. If in the combined first and second sample a total of 5 or more defects are found the lot shall be rejected. The test shall be performed as specified in 4.5.4.1 using test equipment as specified in 4.4.4.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.6 Supersession Data. - This specification includes the requirements of MIL-D-60360 (MU) dated 11 February 1966.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1375-A-137

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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